Games Development

Assignment 02

IT number	Name
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1. Game Title – "Barrier Breaker"

2. Summary of the Game Concept

"Barrier Breaker" is a simple 2D side-scrolling action game where the player controls a driver speeding through a restricted road. Even though the police warn the driver to stop, the player ignores the warnings and continues driving forward.

The objective is to:

- Pass through gaps between "Do Not Cross" police barriers.
- Avoid getting stopped by the police.
- Shoot at incoming police units to survive.
- Collect fuel cans along the road to keep the car moving.

If the player runs out of fuel or gets caught, the game ends. The game focuses on fast movement, timing, and basic shooting mechanic to survive as long as possible.

3. Description of how core features are implemented

Health System

- Player starts with max health 100.
- Hitting obstacles or enemies reduces 20 health .
- A health bar shows current health with a gradient green to red.
- If health reaches '0' game will over.
- Player can pass through objects but still counts as damage.

Scoring System

- Score increases during gameplay.
- When player pass one barrier safely 10 points will added.
- If player shoot one enemy 20 points will add.
- Score resets to 0' when the game restarts.

Health Packs(Fuel)

- Health packs(Fuel) appear on the road.
- Collecting one gives 20 health (max is 100).
- If missed, they disappear automatically

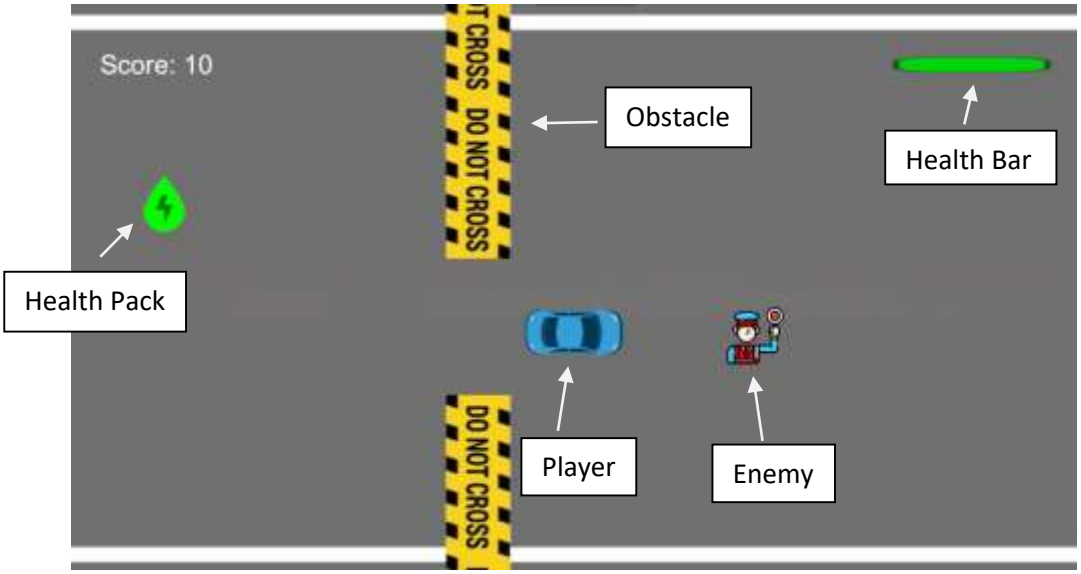
Shooting System

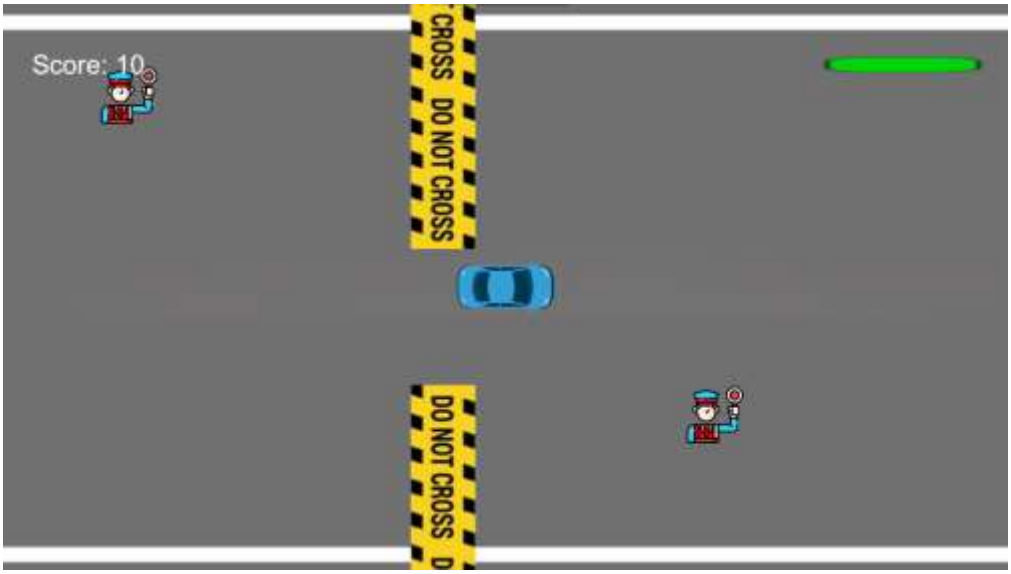
- Player can shoot using mouse click.
- Bullets go towards the mouse direction.
- Bullets simply disappear after hitting something.

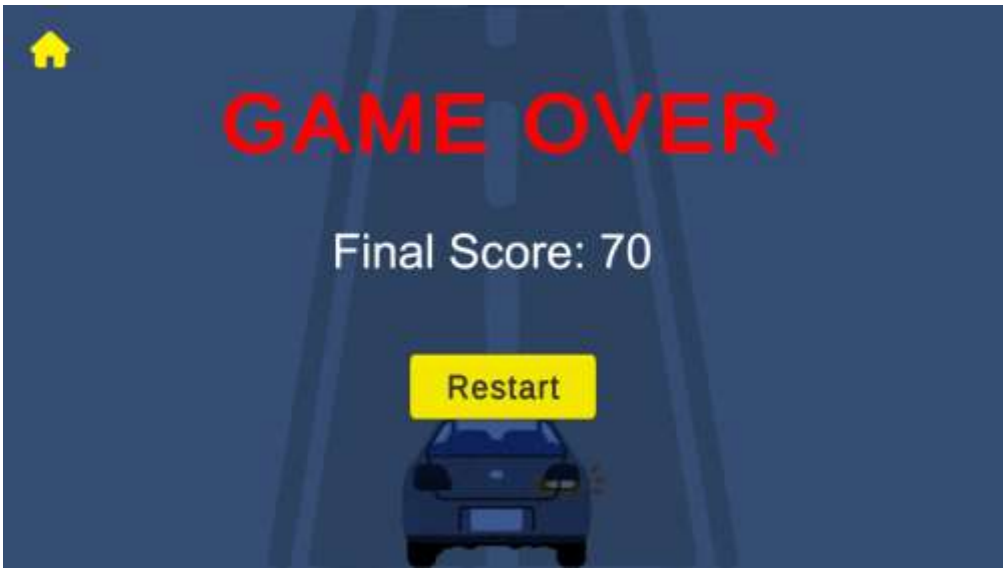
Enemy (Policeman)

- Enemies move toward the player.
- Hitting an enemy reduces player health.
- Shooting enemies gives points (20 points for each).
- Enemies look different from health items.

4. Screenshots of gameplay







5. Control guide (keyboard/mouse input)

Input	Action
W / Up arrow	Move up
S / Down arrow)	Move down
A / D (Left/Right)	Lane adjustment
Mouse left click	Shoot
Restart button	Restart game
Home button	Return to start panel

6. Description of the creative feature

Color of the obstacle(barriers) updates according to the player's health

Police barriers dynamically change color based on the player's current health. This provides an intuitive visual warning of danger, allowing players to have an idea about their health without constantly checking the health bar.

- High health - normal barrier color.
- Low health - barriers transition from light red to dark red.

Dynamic Screen Shake (Impact Feedback)

Screen shake is triggered when the player takes damage, providing immediate visual feedback of impact. This helps the player clearly feel the consequence of hitting obstacles or enemies without relying only on the health bar.

- Triggered when player takes damage.
- Camera shakes briefly on impact.
- Intensity and duration can be adjusted.

Dynamic Gradient Health Bar System

The health bar dynamically changes color based on the player's health percentage, transitioning smoothly from green to yellow to red. This provides quick visual feedback of the player's condition without needing to read values, while also enhancing the UI appearance.

- Color changes based on health percentage.
- Green - high health, Yellow - medium, Red – low.
- Uses currentHealth / maxHealth calculation

7. Credits for any external assets/tools used

Assets from – flaticon.com

8. Video link -

https://drive.google.com/drive/folders/145fUCUOyyoKWbVvN27J6wNsnuoZP4vbO?usp=sharing